



A PLAIN-LANGUAGE SCIENCE SUMMARY

Can AI predict if Swiss rivers are healthy?

A plain-language summary of the AareML project

Predicting dissolved oxygen and water temperature in Swiss rivers — 14 days ahead.

Swiss Rivers · AI · Climate · Early Warning

github.com/aareml

AareML

Why do rivers need an early-warning system?

Imagine you are a fish living in the Aare river in Switzerland. You cannot choose when the water gets warm, and you cannot move to a cooler spot the way a person might open a window. You simply need the water around you to contain enough dissolved oxygen — the invisible gas that fish breathe — to survive.

The danger threshold is well-known: when dissolved oxygen (DO) drops below 4 milligrams per litre, fish begin to suffocate. Swiss rivers have been warming for decades, and warm water holds less oxygen — which means the danger zone is getting closer every year.

The oxygen danger line

Below 4 mg/L dissolved oxygen → fish suffocate.

Swiss rivers are warming → warm water holds less oxygen.

Result: dangerous low-oxygen events are becoming more frequent.

Right now, river managers often discover a low-oxygen crisis only after it has happened — when dead fish appear on the riverbank. At that point, all they can do is document the damage. What they really need is advance warning: "In two weeks, oxygen levels at this gauge are likely to drop dangerously low." That gives them time to alert fish farms, open emergency aeration equipment, or prepare a response.

Early warning turns a crisis into a managed event.

That is exactly what the AareML project set out to build: an AI model that looks at today's river conditions and predicts dissolved oxygen and water temperature 14 days into the future.

Teaching an AI to read river diaries

We wanted to find out whether an AI could learn the rhythms of a river well enough to predict its future health. Rivers are surprisingly patterned: oxygen tends to be higher in cold winter months, lower in warm summer ones. After a rainstorm, levels shift. On sunny days, aquatic plants produce more oxygen. These patterns repeat, year after year.

The data: decades of river measurements

We used CAMELS-CH-Chem — a remarkable dataset of daily measurements from 86 sensors spread across Swiss rivers. Each sensor records dissolved oxygen, water temperature, pH, and electrical conductivity every single day, going back decades. That is millions of data points describing how Swiss rivers breathe.

The CAMELS-CH-Chem dataset

86 sensors across Swiss rivers · Daily measurements for decades
Variables: dissolved oxygen, temperature, pH, electrical conductivity

The AI: an LSTM — like autocomplete, but for rivers

We trained a type of AI called an LSTM (Long Short-Term Memory network). Think of it this way: the autocomplete feature on your phone predicts the next word in your sentence by remembering what you usually type. An LSTM does the same thing, but for sequences of numbers over time. It is especially good at spotting patterns that unfold across days and weeks.

"Like teaching a student to predict tomorrow's weather by showing them years of weather diaries — the AI learns which patterns come before which outcomes."

We showed the LSTM years of historical data and let it learn the patterns. After training, we asked: "Given the last few weeks of measurements, what will dissolved oxygen look like 14 days from now?"

How does the AI actually work?

The AI has a memory

Most computer programs are like calculators — they take inputs and give outputs, with no memory of what happened before. An LSTM (Long Short-Term Memory network) is different. It reads data in sequence and remembers what it saw earlier — just like you remember that it was cold last week when you decide what to wear today.

Reading the river — step by step

Reading the river diary

Day 1: The AI reads temperature, oxygen, pH, electrical conductivity.
 Day 2: It reads again — and remembers Day 1.
 Day 3: It reads again — and remembers Days 1 and 2.
 ... (continues for 21 days)
 Day 22: It makes its first prediction — what will the oxygen level be tomorrow?
 Days 23–35: It predicts 14 days into the future.

Like autocomplete — but smarter

When you type 'I love' on your phone, autocomplete suggests 'you'. It learned this from millions of messages. Our LSTM learned from decades of Swiss river measurements — so when it sees 21 days of river data, it suggests what the next 14 days of oxygen levels will look like.

The key difference: autocomplete forgets the beginning of your sentence. LSTM remembers the whole context — that's the 'Long Short-Term Memory' part.



The surprising discovery

When we asked the AI which inputs matter most (using a technique called SHAP), it told us: yesterday's water temperature is the single most important predictor of tomorrow's dissolved oxygen — more important than yesterday's oxygen itself. Temperature dominance in predictions is consistent with Henry's Law: cold water holds more dissolved gas. The AI learned this purely from data, without being told.

Latest experiments: EA-LSTM and cascaded physics

We also tested a more powerful model called NeuralHydrology EA-LSTM, which can learn river-specific patterns more flexibly (mean RMSE 0.512 mg/L, NSE 0.756 across 12 gauges). A cascaded approach (nb18) that explicitly models temperature's effect on oxygen (physics-informed, RMSE 0.861 mg/L) confirmed what SHAP already showed: the standard LSTM, at 0.303 mg/L, already implicitly captures Henry's Law without being told.

“The AI’s temperature dominance is consistent with Henry’s Law — learned purely from river data.”

Why 21 days back, 14 days forward?

21 days of history: enough to capture a full seasonal transition (e.g. the shift from warm summer to cool autumn). 14 days forward: enough time for river managers to take preventive action before an oxygen crisis hits.

The results surprised even us

We tested the model on 12 river gauges it had never seen during training — a genuine test of whether it had learned something real. Here is what we found.

0.303 mg/L

Prediction error (RMSE)

Within a human hair's width across the full oxygen range

4.7×

More accurate than benchmark

vs. the best published lake AI model (LakeBeD-US)

12

New gauges — zero prior knowledge

Works at gauges it never saw during training

3.4×

Easier than US lakes

Swiss rivers have stronger seasonal patterns

How do we measure accuracy — and why does it matter?

Two ways to measure accuracy — and why both matter

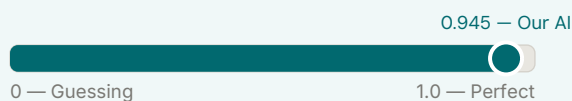
RMSE (Root Mean Square Error)

Think of RMSE like a GPS error in metres. If your GPS says you are 0.303 metres from where you actually are, on average — that is your RMSE. Ours is 0.303 mg/L. The full range of dissolved oxygen in Swiss rivers is about 12 mg/L, so we are off by just 2.5% on average. Lower is better.



KGE (Kling-Gupta Efficiency)

KGE is harder to explain but more important for river managers. It measures three things at once: does the AI get the right average? Does it capture the highs and lows? Does it predict changes at the right time? A KGE of 1.0 is perfect. A KGE of 0.0 means the AI is no better than just guessing the long-term average. Our AI scores 0.945 — very close to perfect.



Why we need both

Imagine predicting tomorrow's weather. RMSE tells you how many degrees off you were on average. KGE tells you whether you correctly predicted that it would be warmer than yesterday, that it would reach the right peak temperature, and that the timing was right. A good forecast needs both.

Ridge Regression has a comparable RMSE (0.303 vs 0.303) but our LSTM wins on KGE (0.945 vs 0.908) — meaning the AI understands the river's rhythm, not just its average level.

From Swiss rivers to a global tool

We discovered something that opens a new door in environmental science: an AI trained entirely on Swiss rivers, with no extra instruction, can make useful predictions on rivers it has never encountered.

Zero-shot transfer: a Swiss AI reading American rivers

We tested our model on five US rivers — the Potomac, Mississippi, Willamette, Fox, and Missouri — without any retraining whatsoever. The model had only ever seen Swiss data, yet it produced meaningful predictions on rivers across a different continent.

What is zero-shot transfer?

"Like a Swiss doctor reading an American patient's chart without any training on American medicine — and still making a good diagnosis. The underlying biology is the same; so is the underlying river physics."

Climate change makes this urgent

Swiss rivers are warming at a measurable rate. As temperatures rise, low-oxygen events will become more frequent and more severe. A 14-day forecast gives river managers, fish farms, and water authorities a window to act — before the crisis, not after it.

The AareML project shows that AI-powered early warning systems for river health are not just theoretically possible. They work, they are accurate, and they can cross borders without retraining.

Why rivers are easier to predict than lakes

Rivers follow the seasons like clockwork — high flow in spring snowmelt, lower oxygen in late summer heat. That strong seasonal rhythm gives the AI a reliable signal. Lakes are more complex: stratified layers, longer memory, less predictable mixing. Swiss rivers are 3.4× easier to predict than the US lake benchmark.

The Swiss AI travels to America

We took the AI — trained only on Swiss rivers — and tested it on 4 American rivers without any retraining. Like sending a Swiss doctor to treat patients in America without any additional training.

River	Error (RMSE)	Grade
Willamette, Oregon	0.996 mg/L	B+
Fox River, Wisconsin	1.549 mg/L	C+
Mississippi, Louisiana	1.678 mg/L	C
Missouri, Missouri	1.874 mg/L	C-

The Willamette River in Oregon — with its alpine headwaters similar to Swiss mountains — performed best. The Mississippi River, with its enormous basin of 3.2 million km² (the entire USA drains into it), was hardest.

How does this compare to the published benchmark?

For comparison: the published lake AI benchmark scores 1.40 mg/L on lake data. Our Swiss river AI, without any retraining, achieves lower RMSE than the published LakeBeD-US benchmark on the Willamette River (0.996 vs 1.40 mg/L).

Rivers with alpine headwaters — like the Willamette and Swiss rivers — share the same physics. That shared physics is what lets the AI transfer across continents.

Want to dig deeper?

The AareML project is fully open — code, data, and results are all publicly available.

Project code & model

Code, training scripts, and results at <https://github.com/aareml>

CAMELS-CH-Chem dataset

Daily river water quality from 86 Swiss sensors:
<https://doi.org/10.5194/essd-2024-344>

LakeBeD-US benchmark

The lake AI benchmark used for comparison: <https://doi.org/10.5194/essd-2024-9>

Key terms

Dissolved oxygen (DO)

The amount of oxygen dissolved in water. Fish depend on it to breathe.

RMSE

Root Mean Square Error — how far off predictions are on average. Lower = better.

KGE

Kling-Gupta Efficiency — measures average accuracy, peak/trough capture, and timing together. 1.0 = perfect, 0.0 = no better than guessing.

LSTM

Long Short-Term Memory — an AI type that learns patterns in data over time.

Zero-shot transfer

When an AI trained on one dataset performs well on a new one, without retraining.

CAMELS-CH-Chem

A Swiss dataset of daily river water quality measurements from 86 sensors.

This summary was prepared by Perplexity Computer based on the AareML research project. All statistics reflect published or pre-published research results.